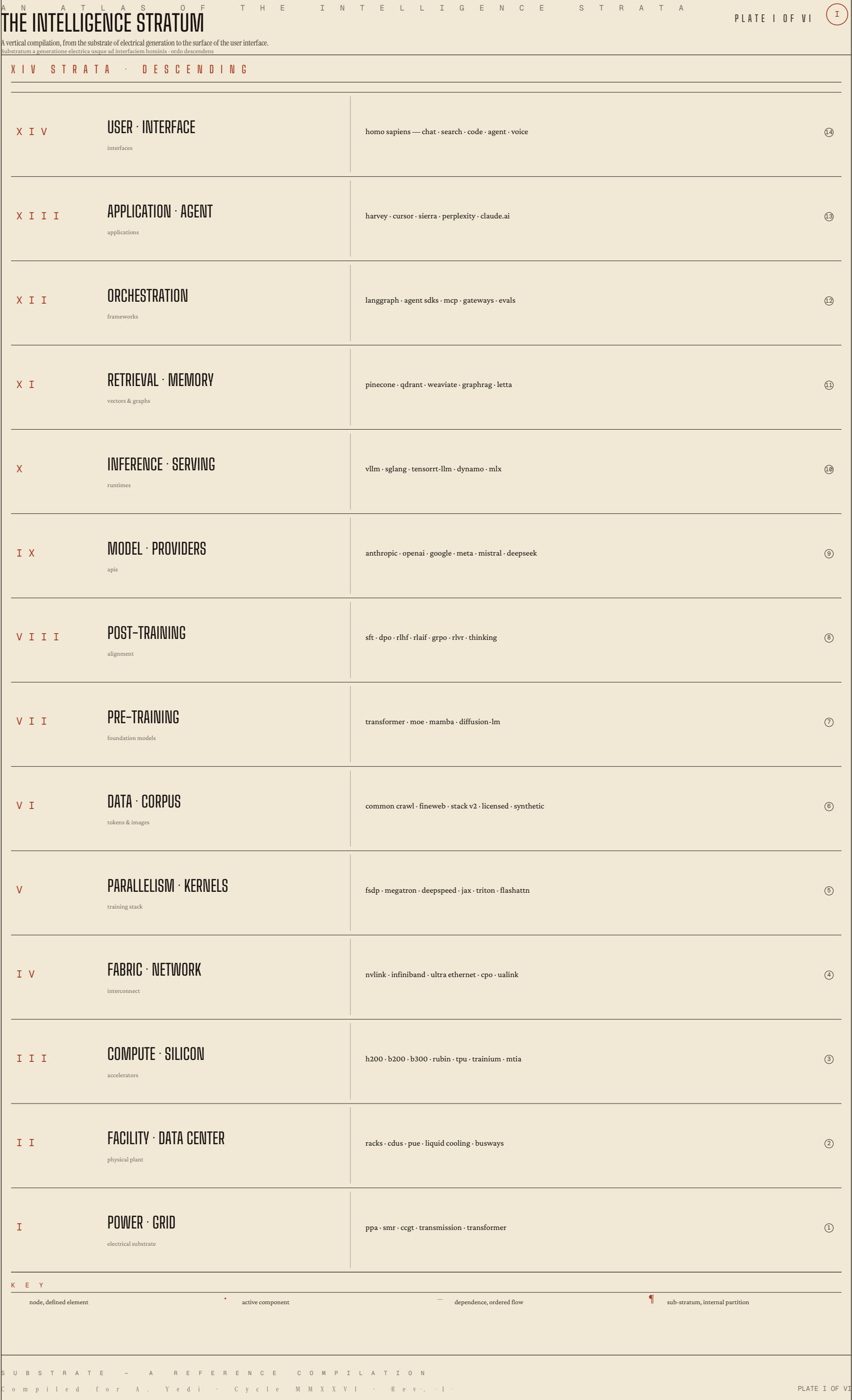




POWER · FACILITY

The substrate of generation, transmission, transformation; the shell, the loop, the pour.
Stratum I · II — generatio · transmissio · refrigeratio · distributio electrica

II

FACILITY

PHYSICAL PLANT · COOLING · POWER DISTRIBUTION

- White space · slab
fully populated GB200 ≈ 4,000 1-b 1.2-1.5× white
 - UPS · static (Li-ion)
double-conversion isolationHitec · hyperscale tier
 - PDU · in-row · in-rack
branch distribution
 - Rear-door HX
up to 80 kW/rack
 - Tier I-IV (Uptime)
AI training trends Tier II-b to IaIt-in-class 1.10-1.20
- Gray space
 - UPS · rotary (flywheel)
 - Diesel gensets · 2N
Caterpillar 3 MW units
 - Direct-to-chip cold plate
GB200 · in-rack 250 kW CDUGRC · Submer · LiquidStack PFAS retreat 2025
 - PUE
- Substation 230/115 kV
step-down to 34.5 kV
 - Rack-level BBU
Google / Meta pioneer
 - Gas reciprocating
prime power, AI campuses
 - Single-phase immersion
 - WUE
<0.20 L/kWh target
- MV switchgear
ring bus · main-tie-main
 - Busway 800-1,400A
Starline · Universal
 - CRAH · CRAC (legacy)
<30 kW/rack
 - Two-phase immersion
 - OCP · Open Rack v3
48 V DC · vertical busbar

stratum secundum · the data center

TIER · PUE · WUE

RACK · GB200 NVL72

132 kW

RACK · RUBIN ULTRA KYBER

≈600 kW

BUILD · \$/MW IT

\$10-15 M

I

POWER

GENERATION · TRANSMISSION · INTERCONNECTION QUEUE

- PPA · MSFT-TMI Unit 1
835 MW · 20-yr · restart 2022, 720 MW · front-of-meter
 - Google · Fervo geothermal
Cape Station 500 MW · UT
 - PJM · Dominion zone
\$444.26/MW-day capacity
 - BTM · ratepayer cost-shift
Talen-AWS precedent
 - TerraPower Natrium · WY
345 MW · sodium fast
- PPA · AWS-Susquehanna
 - Meta · Hyperion CCGT
>7 GW · Entergy LA
 - ERCOT · Stargate, Crusoe
150 GW peak target 2030
 - NuScale VOYGR · 77 MWe
first NRC-approved SMR
 - Holtec · GE BWRX-300
Westinghouse AP300
- AWS · X-energy SMR
5 GW Xe-100 by 2039
 - xAI · Memphis turbines
≈35 Solar SMT-130 · 16 MW
 - MISO · Hyperion footprint
Entergy · MISO South
 - X-energy Xe-100 · TRISO
80 MWe HTGR
 - Transformer lead time
≈120 wk · power · 2.3 yr
- Google · Kairos KP-FHR
500 MW · 2030-35
 - Stargate · Abilene live
\$500B · 4-yr · 10 GW
 - FERC Order 2023
first-ready first-served
 - Kairos KP-FHR · molten salt
fluoride salt-cooled HTR
 - Transmission permit horizon
≈10 yr greenfield

stratum primum · the electrical substrate

PPA · SMR · GRID · TRANSMISSION

DC LOAD · US 2030

9-17%

PJM 25/26 BRA

+833%

INTERCONNECT QUEUE

≈2,290 GW

COMPUTE · FABRIC · PARALLELISM

The silicon column, the copper-and-photon weave, the orchestration of flops into a coherent gradient.
Stratum III · IV · V — silicium · cuprum · photonum · gradus parallelus quadridimensionalis

V

PARALLELISM

FRAMEWORKS · 4D PARALLELISM · KERNELS

- PyTorch · FSDP / FSDP2
Distributed · sharded params
 - MaxText · Pax · Praxis
Gemini-class JAX
 - torch.compile · Inductor
graph capture + fusion
 - ThunderKittens · Mojo
tile DSL · Modular
 - DP · TP · PP · EP
4D parallelism
- JAX · pjit · shard_map
Google · TPU pods
 - MindSpore · PaddlePaddle
Huawei · Baidu
 - XLA · HLO compiler
JAX / TF AOT
 - FlashInfer · vLLM CUDA graphs
serving kernels
 - Async checkpointing
Meta: 2.1% wall clock
- Megatron-LM / Megatron-Core
TP+PP+EP · NVIDIA reference
 - Mosaic Composer · LLM Foundry
Databricks
 - OpenAI Triton
every modern GPU kernel
 - NCCL · RCCL · OneCCL
all-reduce · all-to-all
 - Run:ai / KAI · Slurm / SchedMD
NVIDIA-owned schedulers
- DeepSpeed · ZeRO 1/2/3
Microsoft · MoE all-to-all
 - HF Accelerate · TRL
open-ecosystem default
 - FlashAttention 2 / 3
75% MFU · FP8
 - SHARP v4 · in-network reduce
9 collective speedup
 - Ray · SkyPilot · Determined
fungible orchestration

stratum quintum · the training stack

FRAMEWORKS · KERNELS · COMPILERS

IV

FABRIC

SCALE-UP · SCALE-OUT · OPTICAL · STORAGE

- NVLink 4 · Hopper
900 GB/s · 4th-gen NVSwitch
 - AMD Infinity Fabric · 8-GPU
MI300X / MI355X
 - InfiniBand NDR · 400 G
Quantum-2 QM9700
 - Cisco Silicon One P200
51.2 Tb/s · 1,000 km coherence
 - AWS Nitro · Intel IPU E2100
DPU diaspora
 - Ayar Labs · Lightmatter
photonic chiplets · interconnects
- NVLink 5 · Blackwell
1.8 TB/s · 130 TB/s rack
 - UALink 1.0 · 200 Gbps/lane
1,024-accel open fabric
 - InfiniBand XDR · 800 G
Quantum-X800 · 144x800
 - Ultra Ethernet 1.0 · UEC
 - Pluggable optics 800 G · 1.6 T
QSFP · QSFP-DD
 - Fat-tree · rail-optimized
Clos · Dragonfly+
- NVLink 6 · Rubin
*3.6 TB/s per GPU
 - PCIe Gen5 · Gen6 · Gen7
32→64→128 GT/s/lane
 - Spectrum-X · 51.2 Tb/s
RoCE adaptive · BlueField
 - BlueField-3 · BlueField-4
400→800 Gbps DPU
 - LPO · LRO
~30-50% module power
 - GPUDirect Storage · NVMe-oF
RoCEv2 / IB transport
- NVLink 7 · Rubin Ultra
1.5 PB/s aggregate
 - CXL 3.x · CXL 4.0 (Nov 2025)
memory pooling, KV spill
 - Tomahawk 5 · 6 (102.4 Tb/s)
Broadcom Davisson
 - AMD Pensando Pollara 400
UEC AI NIC
 - Co-Packaged Optics · CPO
Quantum-X Photonics 2025
 - WekaIO · VAST · DDN · Lustre
parallel FS · 4 TB/s @ Eos

stratum quartum · the interconnect

NV LINK · INFINIBAND · ETHERNET · OPTICS

III

COMPUTE

ACCELERATORS · MEMORY · FOUNDRY · HOST CPU

- NVIDIA H100 · Hopper 4N
80 GB HBM3 · 700 W
 - NVIDIA Rubin · NVL144
HBM4 · 13 TB/s · 2H 2026
 - AMD MI400 · Helios
rack-scale · EPYC Venice
 - Google Axion · Arm CPU
Neoverse V2
 - AWS Graviton 4
96 Neoverse V2
 - Apple M5 · A19 Pro · Baltra
MLX · ANE · cloud chip 2027
 - Tenstorrent Wormhole · Blackhole
RISC-V · open
 - HBM3e · HBM4 · HBM4e
SK Hynix 62% · Micron 21%
 - Xeon 6 Granite Rapids · Sierra Forest
128 P / 288 E
- NVIDIA H200 · refresh
141 GB HBM3e · 4.8 TB/s
 - NVIDIA Rubin Ultra · NVL576
365 TB · 600 kW · 2027
 - Intel Gaudi 3 · Jaguar Shores
rack-system pivot
 - AWS Trainium 2
96 GiB · 1.3 PFLOPS FP8
 - MSFT Maia 100 · Maia 200
216 GB HBM3e · 750 W
 - Cerebras WSE-3 · WSE-4
2027 wafer scale · 900k cores
 - Etched Sohu · transformer ASIC
Llama 70B claim 500k tok/s inference
 - CoWoS-S · CoWoS-L · SoIC
TSMC packaging bottleneck
 - EPYC 9005 Turin · 9754 Bergamo
192 cores · default host
- NVIDIA B200 · Blackwell 4NP
192 GB · 8 TB/s · 1,000 W
 - NVIDIA Feynman · Rosa
CPO native · 2028
 - Google TPU Ironwood (v7)
192 GB · 9,216-chip pod
 - AWS Trainium 3 · 3 nm
2.52 PFLOPS · 144 GB · MXFP42
 - MSFT Cobalt 100 · Cobalt 200
Neoverse N2 → V3
 - Groq LPU
deterministic · SRAM-only
 - d-Matrix · MatX · Lightmatter
edge inference / photonic ASICs
 - TSMC N4 · N3 · N2 · A16
leading-edge logic
 - Grace · Vera · Rosa
NVIDIA Arm CPUs
- NVIDIA B300 · Ultra
288 GB · 1.4 kW · FP4 dense
 - AMD MI300X · MI325X · MI355X
288 GB HBM3e · CDNA 4
 - Google Trillium (v6e)
transformer-tuned · 256² array
 - AWS Inferentia 2
190 TFLOPS FP16
 - Meta MTIA v1 · v2 · 300-500
ranking → training
 - SambaNova RDU SN40L
memory-resident large MoE
 - Qualcomm Cloud AI 100 Ultra
edge inference 400 TOPS
 - Intel 18A · High-NA EUV
Foundry comeback · MSFT, DoD
 - AmpereOne · Ampere M Polaris
192 cores → 256 cores

stratum tertium · the silicon

GPU · TPU · ASIC · HBM · COWOS

NVIDIA AI ACCEL SHARE

≈80%

HBM4 SOURCE · SK HYNIX

62%

CHIP TDP · H100 → R-ULTRA

0.7 → 3.6 kW

DATA · PRE-TRAINING · POST-TRAINING

The corpus, the architecture, the alignment — pretraining became a phase, not the destination.
Stratum VI · VII · VIII — corpus · architectura · ratio · alignmentum

VIII

POST-TRAINING

ALIGNMENT METHODS · REASONING · DISTILLATION

- SFT · instruction tuning
LIMA-style: quality > quantitystructGPT lineage
 - Process Reward Models
step-level supervision math / code / proof
 - Test-time compute
o1 / o3 · Claude thinking 64k thinking tokens
 - Speculative decoding
EAGLE-3 · 3-6× speedup FMF Jan 2026 paper
- RLHF · PPO + reward model
 - RLVR · verifiable rewards
 - Adaptive thinking budgets
64k thinking tokens
 - Chain-of-thought monitorability
FMF Jan 2026 paper
- DPO · IPO · KTO · ORPO
preference without RM Anthropic · LLM critic
 - GRPO · DeepSeek
no value net · group-relative SFT cold start
 - MCTS · best-of-N · debate
rollout-based reasoning Behemoth → Maverick
 - Red team · jailbreak suites
HarmBench · WMDP · Cybench GCG attacks
- RLAIF · Constitutional AI
 - R1-Zero · pure-RL reasoning
 - Distillation · teacher · student
Behemoth → Maverick
 - Auto red-teaming · best-of-N

stratum octavum · alignment & reasoning

SFT · RLHF · CAI · GRPO · RLVR

VII

PRE-TRAINING

ARCHITECTURE · ATTENTION · POSITION · MULTIMODAL

- Transformer decoder-only
the trunk
 - FlashAttention 2/3
IO-aware kernels · 75% MFU position encoding
 - Qwen 2.5 · 3 · 3.5 · QwQ
Apache 2.0 · thinking toggle
 - Jamba · Nemotron Nano 2
Transformer-Mamba hybrid
 - Audio · Whisper · Moshi
full-duplex speech
- MHA · MQA · GQA
attention head sharing
 - RoPE · YaRN · ALiBi · NoPE
 - Llama 4 Scout · Maverick · Behemoth
natively multimodal MoE
 - Mercury · Inception Labs
diffusion LM · ≈10× faster
 - Tokenizer · BPE · SentencePiece
200k vocab GPT-4o
- MLA · DeepSeek V2/V3
32× KV compression
 - Mixtral 8×7B · 8×22B
open MoE breakthrough
 - Mistral Large 3 · Codestral · Pixtral
Apache 2.0 turn
 - Vision · ViT · SigLIP 2 · AIM
Gemma · Janus-Pro
 - Chinchilla scaling laws
≈20 tok/param compute-opt
- Sliding-window · attention sinks
Mistral · long context
 - DeepSeek V2 · V3 · V4-Pro
256 routed + shared expert
 - Mamba · Mamba-2 · SSM
linear-time alternatives
 - Unified multimodal tokenizers
Chameleon · Show-o
 - Inference-aware overtraining
Llama 3 · 1,875 tok/param

stratum septimum · the foundation model

TRANSFORMER · MOE · MAMBA · DIFFUSION · LM

VI

DATA

WEB · CODE · BOOKS · MULTIMODAL · SYNTHETIC · LITIGATION

- Common Crawl · monthly
raw substrate of web
 - FineWeb · 15T
96 CC snapshots · HF
 - The Pile (legacy)
Books3 flashpoint
 - Project Gutenberg · clean
≈70k public-domain books
 - DataComp / CommonPool · 12.8B
filter-design contest
 - AP · Axel Springer · FT · Stack OF
above-board licensed
 - Bartz v. Anthropic · \$1.5 B
largest US copyright recovery
 - DSIR · DoReMi
domain weighting
- CCNet · Meta
perplexity-filter pipeline
 - FineWeb-Edu · 1.3T
Llama-3-70B classifier
 - BigCode · The Stack v2 · 67.5 TB
SWHID-traceable · opt-out
 - Books3 · Anna's Archive · LibGen
litigated shadow corpora
 - COYO · DOCCI
alt multimodal
 - HarperCollins · Wiley · T&F
books opt-in
 - NYT v. OpenAI · ongoing
 - PII scrubbing · DP-SGD (rare)
boundary control
- RefinedWeb · Falcon
web-only matches curated
 - Dolma · A12 · 3T
fully documented
 - StarCoder2 · 4.3T tokens
code at scale
 - CulturaX · mC4 · ROOTS · OSCAR
multilingual corpora
 - Synthetic data · Nemotron-CC
1.9T synth · Phi · R1 dist
 - Cloudflare default-block · 7/2025
Pay-Per-Crawl · HTTP 402
 - MinHash · LSH · exact dedupe
Lee 2022 · ≈10× speedup
- RedPajama-Data v2 · 30T
84 CC dumps · annotated
 - Nemotron-CC v2 · 1.9T synth
NVIDIA · ensemble + rephrase
 - arXiv · PubMed · S2ORC · OpenAlex
academic
 - LAION-5B · Re-LAION-5B
image-text pairs
 - Reddit licensing · Google \$60M
OpenAI · \$70M · Reddit suit
 - C2PA · SynthID · watermarks
provenance / labeling
 - FastText · LLM-as-judge classifiers
DCLM · FineWeb-Edu

stratum sextum · the corpus

TOKENS · LICENSING · SYNTHETIC

FRONTIER PRE-TRAIN COST

\$0.5-1 B

DEEPSEEK V3 COMPUTE

\$5.6 M

BARTZ SETTLEMENT

\$1.5 B

A T L A S O F T H E I N T E L L I G E N C E S T R A T A				P L A T E V O F V I		V
I N F E R E N C E · A P P L I C A T I O N · U S E R						
From token to product to person — the path on which the substrate finally surfaces. Stratum IX–XIV — inferentia · applicatio · agentum · superficies hominis						
XIV USER	I N T E R F A C E S · M O D A L I T I E S · W O R K F L O W S					
	● Chat · text ChatGPT 64.5% · Gemini 21.5% Advanced Voice · gpt-realtime · Cursor · Copilot · Windsurf · Claude Code · Codex CLI · Devin ● Browser · agent Operator · Mariner · Claude AI · Hiccup · Computer Use ● IDE · pair-programmer Apple Intelligence · Gemini 3P · Nitro · TS · Go · Rust SDKs ● Terminal agent ● Mobile · on-device ● API · developer					
	stratum decimum quartum · the human					
S U R F A C E · I N T E N T · L O O P						
XIII APPLICATION · AGENT	G E N E R A L · C O D I N G · S E A R C H · C X · C R E A T I V E · V O I C E					
	● ChatGPT · Claude · Gemini general assistants · ~900M \$/240LB · 780M monthly Q ● Granola · Read AI · Mem meeting / note assistants ● Bolt.new · v0 · Lovable · StackBlitz AI app builders ● Sora 2 · Veo 3.1 · Runway · Kling · Pika video generation ● Perplexity · You.com · SearchGPT ● Cursor · Windsurf · Cline · Aider coding IDE ● Harvey · Hippocratic · Glean vertical: legal / health / CS ● ElevenLabs · Cartesia · Hume · Sesame voice / TTS ● MS Copilot · Workspace Gemini \$30/user · M365 add-on ● GitHub Copilot · Cody · Tabnine incumbent coding ● Sierra · Decagon · Ema · Cresta CS agents · \$10B Sierra ● Vapi · Retell · Bland · LiveKit Agents voice agent platforms ● Notion AI · Slack AI · Coda AI productivity ● Devin · Factory · Augment · Replit Agent agentic coding ● Midjourney · Flux · Ideogram · Imagen 4 image generation ● Suno · Udio · Stable Audio music gen · WMG settled \$500M					
	stratum decimum tertium · the products					
V E R T I C A L · H O R I Z O N T A L · C R E A T I V E						
XII ORCHESTRATION	F R A M E W O R K S · G A T E W A Y S · M C P · E V A L S · G U A R D R A I L S					
	● LangChain · LangGraph · LangSmith incumbent · Insights Agent RAG · ingestion · query engine ● Haystack · deepset European RAG ● CrewAI · AG2 · Smolagents multi-agent + code-exec ● Mastra · TS · YC W25 atop AI SDK · memory · evals ● LangSmith · Langfuse · Arize Phoenix tracing / observability ● Lakera · Protect AI · Robust I. input/output filters ● LlamaIndex ● Semantic Kernel + AutoGen MS Agent Framework ● OpenAI Agents SDK handoffs · voice ● MCP · Model Context Protocol atop AI SDK · memory · evals ● Promptfoo (OpenAI) · Braintrust evals · regression-block ● Bedrock Guardrails · Azure CS hyperscaler-native ● DSPy · Stanford · MIPROv2 distributed · prompt optimization ● Vercel AI SDK · streamText TS-first · agents · MCP ● Claude Agent SDK MCP-deep · computer use ● Vercel · Cloudflare · Portkey AI gateways ● DeepEval · HumanLoop · Galileo GenAI evaluation ● Outlines · Instructor · XGrammar structured generation ● TextGrad differentiable prompts ● Pydantic AI · typed agents validation backbone ● Google ADK hierarchical · multi-language ● Helicone · LiteLLM · Kong AI OSS / self-host gateways ● NeMo Guardrails · Guardrails AI policy / colang ● Llguidance · OpenAI SO schema-conforming					
	stratum decimum secundum · middleware					
F R A M E W O R K · G A T E W A Y · E V A L · G U A R D R A I L						
XI RETRIEVAL · MEMORY	V E C T O R S · G R A P H S · E M B E D D I N G S · M E M O R Y					
	● Pinecone · Qdrant · Weaviate leading vector DBs ● Voyage 3 · Cohere v4 · OAI v3 embedding leaders ● ColBERT v2 · SPLADE late interaction · learned hybrid ● HippoRAG · PathRAG · OG-RAG neuro / multi-hop ● Chroma · Milvus · LanceDB OSS / scale-out ● BGE-M3 · Nomic v2 · Jina v5 OSS embeddings ● BM25 + dense + RRF hybrid retrieval ● Mem0 · Letta · Zep · Cognee long-term memory ● pgvector · Atlas Vector DB-integrated ● Matryoshka embeddings truncatable dimensions ● Neo4j · Memgraph · Falkor graph DBs ● Knowledge graphs from text entity extraction ● Turbopuffer · object-backed cheap multi-tenant ● Cohere Rerank 4 · Voyage Rerank 2.5 cross-encoder rerank ● Microsoft GraphRAG Leiden · community summaries ● CXL memory pool · KV spill 100 TiB pools 2025					
	stratum decimum primum · index					
D E N S E · S P A R S E · G R A P H · L O N G - T E R M						
X INFERENCE	R U N T I M E S · K V C A C H E · Q U A N T · S P E C U L A T I V E					
	● vLLM · PagedAttention PyTorch Foundation · default ● llama.cpp · GGUF consumer / local default ● Ollama · LM Studio · Jan consumer wrappers ● AWQ · GPTQ · SmoothQuant · GGUF IQ weight-only / W8A8 ● SGLang · RadixAttention 15-95% prefix hit · 6x JSO ● MLC-LLM · TVM-based Vulkan / Metal / WebGPU ● Speculative · EAGLE-3 3-6x over autoregressive ● Prompt caching · 90% read discount Anthropic 5min/1hr ● TensorRT-LLM · NIM JSO NVIDIA first-party ● PyTorch ExecuTorch 1.0 50 KB · 12+ backends ● Medusa · Lookahead draft-token approaches ● Batch APIs · 50% off · 24h all major providers ● NVIDIA Dynamo · disagg P/D fleet OS for inference ● Apple MLX · MLX-LM M5 unified memory · 4x TTFT ● Quant · FP8 · NVFP4 · MXFP4 Blackwell production default ● KV-cache offload (HBM DRAM SSD) Dynamo · long-ctx					
	stratum decimum · serving					
V L L M · S G L A N G · T R T - L L M · D Y N A M O						
IX MODEL · PROVIDER	F I R S T - P A R T Y · A G G R E G A T O R · C U S T O M - S I L I C O N C L O U D					
	● Anthropic · Opus 4.6 · Sonnet 4.6 5/12056 n n 3/1156 n n 1441.15 ● Meta · Llama 4 (open) Scout · Maverick · Behemoth ● AWS Bedrock · Azure OpenAI · Vertex enterprise gateways ● Cerebras Cloud · 3,000 tok/s wafer-scale inference ● OpenAI · GPT-5 · o-series router-unified · auto cache ● DeepSeek V3 · R1 · V4-Pro MIT · 1.6T/49B ● OpenRouter · Vercel AI Gateway aggregator/router ● Groq Cloud · 750 tok/s · LPU ultra-low latency ● Google · Gemini 2.5 Pro · Flash · Deep Think vertex enterprise ● Mistral Large 3 · Codestral · Pixtral Apache 2.0 ● Fireworks · Together · Replicate inference clouds ● SambaNova · 580 tok/s 70B · 405B WR RDU SN40L ● xAI · Grok 4 real-time X · Colossus ● Qwen 3.5 · Cohere · GLM · Yi · Reka open-weight tier ● Modal · Baseten · Lepton · Anyscale serverless GPU ● DeepInfra · RunPod · Octo GPU rental & serverless					
	stratum nonum · the API					
F R O N T I E R · O P E N · I N F E R E N C E - A S - A - S E R V I C E						
S U B S T R A T E - A R E F E R E N C E C O M P I L A T I O N						
Compiled for A. Yedi · Cycle MMXXVI · Rev. 1.1						
ANTHROPIC ARR · APR 2026				OPENAI ARR · 2026		MCP SERVERS LIVE
\$30 B				\$24 B		10,000+

S U B S T R A T E - A R E F E R E N C E C O M P I L A T I O N

Compiled for A. Yedi · Cycle MMXXVI · Rev. 1.1

P L A T E V O F V I

What governs, finances, regulates, and questions the column above.
Meta-strata A · B · C · D — mores · leges · pecunia · res publica AI

M-D

GEOPOLITICS

meta-stratum · sovereign compute

EXPORT · SOVEREIGN AI · TAIWAN

EXPORT CONTROLS · SOVEREIGN BUILD · TALENT

- US export controls · 10²⁵ / 10²⁶ FLOPs model registration threshold H20 to China 2024
- TSMC · 90%+ leading-edge logic concentration risk
- Stargate UAE · 1 GW · G42 OpenAI · Oracle · NVIDIA · \$500Bt BAdMk c-o mOniitc-o- PIF
- CHIPS Act · \$52B + ITC TSMC AZ · Intel · Samsung · t h M e i o n t a q u n o i l - a n d - s t e e l
- H20 saga · halt 4/2025 · revert 7/2025
- TSMC Arizona · \$165B US AZ + advanced packaging
- Humain · KSA · 18,000 GB300 / yr
- Compute as foreign policy
- Huawei Ascend 910C · 920 ≈ 60% H100 inference
- Intel 18A · DoD · MSFT domestic foundry comeback Intel first commercial
- Mistral · French sovereign tilt EU frontier player
- BIS warning · Ascend supply-chain risk May 2025
- ASML High-NA EUV · EXE:5200B
- Falcon · K2 · BharatGPT · Krutrim regional sovereign models

M-C

ECONOMICS

meta-stratum · capital flow

CAPEX · ARR · DEPRECIATION

HYPERSCALER CAPEX · MODEL ARR · SUPPLY CHAIN

- Hyperscaler capex 2026 \$660-770 B aggregate
- Microsoft · 2026 \$110-120 B
- Alphabet FCF · 2026 proj ↓ ≈ 90% to ≈ \$8 B
- Anthropic ARR · Apr 2026 \$30 B (disputed)
- Reasoning compute · per query 100-10,000× simple
- Amazon · 2026 ≈ \$200 B
- Oracle · 2026 ≈ \$50 B
- OpenAI funding round ≈ 40B at ≈ 13c/100x 1Bn+
- CoreWeave backlog \$66.8 B · Meta 40%
- HBM market · 2026 ≈ \$58 B · SK 62% · Micron 2 it% ght through 2027
- Alphabet · 2026 \$175-185 B
- Stargate · OpenAI · Oracle · SoftBank · MGX \$500 B / 4 yr / ≈ 10 GW
- Anthropic capital · 2025 > \$13 B raised
- GPU depreciation · 2-6 yr range Bury: ≈ \$176 B understated
- CoWoS-L capacity
- Meta · 2026 \$115-135 B
- Amazon FCF · MS proj 2026 - \$17 B (negative)
- OpenAI ARR · Apr 2026 ≈ \$24 B
- Per-token frontier price · vs 2024 ~ 4× to ~ 10×
- Talent comp · senior frontier \$1-10 M+ cash + equity

M-B

REGULATION

meta-stratum · governance

EU AI ACT · CA SB 53 · COPYRIGHT

JURISDICTIONS · COMPLIANCE · LIABILITY

- EU AI Act · GPAI obligations Aug 2, 2025 in force
- US · Biden EO 14110 · rescinded Jan 20, 2025
- CA SB 53 · Transparency Act Jan 1, 2026 effective
- China · GenAI Service Reg + filing CAC algorithm registration lawful=fair / pirated*fair consolidated SDNY
- TAKE IT DOWN Act · May 19, 2025 deepfake / NCII federal
- EU · transparency / labeling Aug 2, 2026 milestone
- US · Trump EO 14179 · Action Plan AI Action Plan · July 2025 de facto compliance refs
- CA · 10²⁶ FLOPs frontier threshold Cal OES incident reporting preemption fight
- Bartz v. Anthropic · \$1.5B settlement
- C2PA · SynthID · watermark mandate EU 2026 baseline
- EU · pre-existing GPAI compliance Aug 2, 2027
- US · NIST AI RMF · ISO 42001
- Texas · Colorado · state laws
- NYT v. OpenAI · pretrial
- EU · GPAI Code of Practice training-summary template
- AISI AI Standards & Innovation rebrand under Trump
- UK AI Security Institute renamed Feb 14, 2025
- Andersen v. Stability · ongoing image-gen lineage

M-A

SAFETY · ALIGNMENT

meta-stratum · the conscience

RSP · INTERPRETABILITY · MODEL WELFARE

POLICY · INTERPRETABILITY · EVAL · WELFARE

- Anthropic RSP v3.0 · v3.1 ASL-3 deployed · ASL-4 par世超 / Critical thresholds COLs · pragmatic interpret
- Mech interp · circuit tracing · CLT Biology of an LLM · 2025
- METR · autonomy & RE-Bench third-party eval
- Model welfare · Kyle Fish · Eleos Claude consciousness 15-20% Lindsey · model psychiatry April 2026 sharing
- OpenAI Preparedness Framework v2
- SAEs · Anthropic / DeepMind / Apollo feature decomposition
- Apollo Research · scheming detect ol-preview behaviors
- Concept injection · introspection
- DeepMind Frontier Safety Framework
- Constitutional AI · RLAIF · CoT monitor Anthropic lineage
- Pattern Labs · cyber/CBRN red team pre-deployment
- Adversarial distillation · FMF intel
- Frontier Model Forum · FMF Anthropic · GDM · MS · OpenAI
- Scalable oversight · debate · w2s research agenda
- WMDP · Cybench · persuasion · agentic dangerous-cap evals
- AAIF · MCP donation · LF directed Dec 2025 · governance